

Binary & Decimal Number Systems

Practice Set

1**Convert the decimal number 217 to its binary equivalent.**

Solution

Step 1 — Successive Division by 2

Divide 217 repeatedly by 2 and record the remainder at each step:

$$217 \div 2 = 108, \text{ remainder} = 1$$

$$108 \div 2 = 54, \text{ remainder} = 0$$

$$54 \div 2 = 27, \text{ remainder} = 0$$

$$27 \div 2 = 13, \text{ remainder} = 1$$

$$13 \div 2 = 6, \text{ remainder} = 1$$

$$6 \div 2 = 3, \text{ remainder} = 0$$

$$3 \div 2 = 1, \text{ remainder} = 1$$

$$1 \div 2 = 0, \text{ remainder} = 1$$

Step 2 — Read Remainders Bottom-to-Top

Collecting remainders from last to first gives: Binary = **11011001**

Step 3 — Verification

$$\begin{aligned} &1 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 128 + 64 + 0 + 16 + 8 + 0 + 0 + 1 = 217 \end{aligned}$$

Answer: $(217)_{10} = (11011001)_2$

2**Convert the binary number 10110110 to its decimal equivalent.****Solution****Step 1 — Write Position Values (right to left, starting from 0)**

Bit positions: 7 6 5 4 3 2 1 0

Binary digits: 1 0 1 1 0 1 1 0

Step 2 — Multiply Each Bit by 2 raised to its Position

$$1 \times 2^7 = 128, 0 \times 2^6 = 0, 1 \times 2^5 = 32, 1 \times 2^4 = 16$$

$$0 \times 2^3 = 0, 1 \times 2^2 = 4, 1 \times 2^1 = 2, 0 \times 2^0 = 0$$

Step 3 — Sum All Products

$$128 + 0 + 32 + 16 + 0 + 4 + 2 + 0 = 182$$

Answer: $(10110110)_2 = (182)_{10}$ **3****In the binary number 11010, find the place value and face value of every digit. Also compute the sum of all place values.****Solution****Step 1 — Definitions****Face Value** — The digit itself (always 0 or 1 in binary).**Place Value** — Face Value $\times 2^{\text{position}}$ (position counted from right, starting at 0).**Step 2 — Compute for Each Digit (right to left)**

$$\text{Position 0, digit 0: Face Value} = 0, \text{Place Value} = 0 \times 2^0 = 0$$

$$\text{Position 1, digit 1: Face Value} = 1, \text{Place Value} = 1 \times 2^1 = 2$$

$$\text{Position 2, digit 0: Face Value} = 0, \text{Place Value} = 0 \times 2^2 = 0$$

$$\text{Position 3, digit 1: Face Value} = 1, \text{Place Value} = 1 \times 2^3 = 8$$

$$\text{Position 4, digit 1: Face Value} = 1, \text{Place Value} = 1 \times 2^4 = 16$$

Step 3 — Sum of All Place Values

$$0 + 2 + 0 + 8 + 16 = 26, \text{ which equals the decimal value of } (11010)_2.$$

Answer: Place Values: 0, 2, 0, 8, 16. Sum of Place Values = 26 = $(11010)_2$ in decimal.

4

In the decimal number 4096, find the place value of each digit. Then verify that $4096 = 2^{12}$ by converting it to binary.

Solution

Step 1 — Place Values in Decimal (Base 10)

Position 3, digit **4**: Place Value = $4 \times 10^3 = 4000$

Position 2, digit **0**: Place Value = $0 \times 10^2 = 0$

Position 1, digit **9**: Place Value = $9 \times 10^1 = 90$

Position 0, digit **6**: Place Value = $6 \times 10^0 = 6$

Verification: $4000 + 0 + 90 + 6 = 4096$

Step 2 — Convert 4096 to Binary by Successive Division

$4096 \div 2 = 2048$, remainder = **0**

$2048 \div 2 = 1024$, remainder = **0**

$1024 \div 2 = 512$, remainder = **0**

$512 \div 2 = 256$, remainder = **0**

$256 \div 2 = 128$, remainder = **0**

$128 \div 2 = 64$, remainder = **0**

$64 \div 2 = 32$, remainder = **0**

$32 \div 2 = 16$, remainder = **0**

$16 \div 2 = 8$, remainder = **0**

$8 \div 2 = 4$, remainder = **0**

$4 \div 2 = 2$, remainder = **0**

$2 \div 2 = 1$, remainder = **0**

$1 \div 2 = 0$, remainder = **1**

Binary = **1000000000000** (1 followed by twelve 0s, confirming 2^{12})

Answer: Decimal place values: 4000, 0, 90, 6. $(4096)_{10} = (1000000000000)_2 = 2^{12}$

5

A computer system uses 8-bit memory addresses. (a) How many unique memory locations can it address? (b) What is the highest address in decimal? (c) What is the address of the 200th memory location in binary?

Solution**Part (a) — Total Addressable Locations**

With 8 bits, each bit can be 0 or 1, giving 2^8 combinations.

$2^8 = 256$ **unique memory locations** (addresses 0 through 255).

Part (b) — Highest Address in Decimal

The highest address has all bits set to 1: $(11111111)_2$

$$= 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$

$$= 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$$

Part (c) — 200th Memory Location in Binary

Addresses start from 0, so the 200th location has address **199**.

$$199 \div 2 = 99, \text{ remainder} = 1$$

$$99 \div 2 = 49, \text{ remainder} = 1$$

$$49 \div 2 = 24, \text{ remainder} = 1$$

$$24 \div 2 = 12, \text{ remainder} = 0$$

$$12 \div 2 = 6, \text{ remainder} = 0$$

$$6 \div 2 = 3, \text{ remainder} = 0$$

$$3 \div 2 = 1, \text{ remainder} = 1$$

$$1 \div 2 = 0, \text{ remainder} = 1$$

Binary = **11000111**

Answer: (a) 256 locations (b) 255 (c) $(11000111)_2$

6

In an RGB color model, each channel (Red, Green, Blue) uses 8 bits. A pixel has R=200, G=130, B=75. (a) Convert each channel value to binary. (b) How many total bits are needed per pixel? (c) How many distinct colors can be represented?

Solution

Part (a) — Convert Red = 200 to Binary

$$200 \div 2 = 100, \text{ remainder} = 0$$

$$100 \div 2 = 50, \text{ remainder} = 0$$

$$50 \div 2 = 25, \text{ remainder} = 0$$

$$25 \div 2 = 12, \text{ remainder} = 1$$

$$12 \div 2 = 6, \text{ remainder} = 0$$

$$6 \div 2 = 3, \text{ remainder} = 0$$

$$3 \div 2 = 1, \text{ remainder} = 1$$

$$1 \div 2 = 0, \text{ remainder} = 1$$

$$\text{Binary} = \mathbf{11001000}$$

Convert Green = 130 to Binary

$$130 \div 2 = 65, \text{ remainder} = 0$$

$$65 \div 2 = 32, \text{ remainder} = 1$$

$$32 \div 2 = 16, \text{ remainder} = 0$$

$$16 \div 2 = 8, \text{ remainder} = 0$$

$$8 \div 2 = 4, \text{ remainder} = 0$$

$$4 \div 2 = 2, \text{ remainder} = 0$$

$$2 \div 2 = 1, \text{ remainder} = 0$$

$$1 \div 2 = 0, \text{ remainder} = 1$$

$$\text{Binary} = \mathbf{10000010}$$

Convert Blue = 75 to Binary

$$75 \div 2 = 37, \text{ remainder} = 1$$

$$37 \div 2 = 18, \text{ remainder} = 1$$

$$18 \div 2 = 9, \text{ remainder} = 0$$

$$9 \div 2 = 4, \text{ remainder} = 1$$

$$4 \div 2 = 2, \text{ remainder} = 0$$

$$2 \div 2 = 1, \text{ remainder} = 0$$

$$1 \div 2 = 0, \text{ remainder} = 1$$

$$\text{Binary} = \mathbf{01001011} \text{ (padded to 8 bits with a leading zero)}$$

Part (b) — Total Bits per Pixel

$$8 \text{ bits (R)} + 8 \text{ bits (G)} + 8 \text{ bits (B)} = \mathbf{24 \text{ bits per pixel}}$$

Part (c) — Total Distinct Colors

Each channel has $2^8 = 256$ possible values (0 to 255).

Total colors = $256 \times 256 \times 256 = 2^{24} = 16,777,216$

Answer: $R = (11001000)_2$, $G = (10000010)_2$, $B = (01001011)_2$. Bits/pixel = 24. Distinct colors = 16,777,216

7

The binary number $A = 10100100$. (a) List all positions where the digit 1 appears and compute the place value at each. (b) A new binary number B is formed by flipping every bit of A (each 0 becomes 1 and each 1 becomes 0). Find the decimal value of B. (c) What is the sum of the decimal values of A and B?

Solution**Part (a) — Place Values of 1-bits in $A = 10100100$**

Positions (right to left): 7 6 5 4 3 2 1 0

Digits: 1 0 1 0 0 1 0 0

The digit 1 appears at positions 2, 5, and 7.

Position 2: $1 \times 2^2 = 4$

Position 5: $1 \times 2^5 = 32$

Position 7: $1 \times 2^7 = 128$

Decimal value of A = $4 + 32 + 128 = 164$

Part (b) — Flip All Bits to Get B

$A = 1\ 0\ 1\ 0\ 0\ 1\ 0\ 0$

$B = 0\ 1\ 0\ 1\ 1\ 0\ 1\ 1$ (every bit inverted)

Decimal value of B:

$$0 \times 2^7 + 1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 0 + 64 + 0 + 16 + 8 + 0 + 2 + 1 = 91$$

Part (c) — Sum of A and B

$$A + B = 164 + 91 = 255$$

This is not a coincidence. Flipping all bits of an 8-bit number and adding it to the original always gives 11111111 in binary, which equals $2^8 - 1 = 255$.

In general, for an n-bit number: $A + \text{flip}(A) = 2^n - 1$.

This property is fundamental to how computers represent negative numbers using one's complement.

Answer: (a) Place values: 4, 32, 128; A = 164 (b) $B = (01011011)_2 = 91$ (c) $A + B = 255 = 2^8 - 1$

8

A mystery 8-bit binary number N satisfies all of the following conditions: (i) The sum of place values of the leftmost 4 bits equals 192. (ii) The sum of place values of the rightmost 4 bits equals 11. (iii) The face value of the bit at position 4 is 0. Find N in binary, then convert it to decimal.

Solution

Step 1 — Understand Bit Layout

An 8-bit number has positions 7, 6, 5, 4 (leftmost 4) and 3, 2, 1, 0 (rightmost 4).

The place value weights are:

Left: $2^7 = 128$, $2^6 = 64$, $2^5 = 32$, $2^4 = 16$

Right: $2^3 = 8$, $2^2 = 4$, $2^1 = 2$, $2^0 = 1$

Step 2 — Solve Constraint (iii) First

Position 4 has face value 0, so bit 4 = 0. Its place value contribution is 0.

The leftmost 4 bits are at positions 7, 6, 5, 4. Since bit 4 = 0, we need positions 7, 6, 5 alone to sum to 192.

Step 3 — Solve Constraint (i): Left Sum = 192

Available weights: 128, 64, 32 (position 4 is already 0).

$128 + 64 = 192$. This accounts for the full sum.

So bit 7 = 1, bit 6 = 1, bit 5 = 0, bit 4 = 0.

Step 4 — Solve Constraint (ii): Right Sum = 11

Available weights for positions 3, 2, 1, 0: 8, 4, 2, 1.

We need a combination summing to 11.

$8 + 2 + 1 = 11$. So bit 3 = 1, bit 2 = 0, bit 1 = 1, bit 0 = 1.

Step 5 — Assemble and Verify

N = 11001011

Decimal = $128 + 64 + 0 + 0 + 8 + 0 + 2 + 1 = 192 + 11 = 203$

Left 4 bits (1100): $128 + 64 + 0 + 0 = 192$

Right 4 bits (1011): $8 + 0 + 2 + 1 = 11$

Bit at position 4 = 0. All three constraints satisfied.

Answer: $N = (11001011)_2 = (203)_{10}$

9

Two binary numbers $A = 110111011$ and $B = 101100110$ represent scores of two players. (a) Convert both to decimal. (b) Find the decimal difference $A - B$. (c) Convert this difference back to binary. (d) What is the sum of the place values of all 1-bits in the binary form of the difference?

Solution

Part (a) — Convert A and B to Decimal

$A = 110111011$ (9 bits, positions 8 to 0)

$$1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ = 256 + 128 + 0 + 32 + 16 + 8 + 0 + 2 + 1 = 443$$

$B = 101100110$ (9 bits, positions 8 to 0)

$$1 \times 2^8 + 0 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ = 256 + 0 + 64 + 32 + 0 + 0 + 4 + 2 + 0 = 358$$

Part (b) — Decimal Difference

$$A - B = 443 - 358 = 85$$

Part (c) — Convert 85 to Binary

$$85 \div 2 = 42, \text{ remainder} = 1$$

$$42 \div 2 = 21, \text{ remainder} = 0$$

$$21 \div 2 = 10, \text{ remainder} = 1$$

$$10 \div 2 = 5, \text{ remainder} = 0$$

$$5 \div 2 = 2, \text{ remainder} = 1$$

$$2 \div 2 = 1, \text{ remainder} = 0$$

$$1 \div 2 = 0, \text{ remainder} = 1$$

$$\text{Binary} = 1010101$$

Part (d) — Sum of Place Values of All 1-bits in 1010101

Positions with digit 1: position 0, 2, 4, 6

$$\text{Place values: } 2^0 = 1, 2^2 = 4, 2^4 = 16, 2^6 = 64$$

$$\text{Sum} = 1 + 4 + 16 + 64 = 85 \text{ (equals the decimal value itself, as expected)}$$

Answer: (a) $A = 443$, $B = 358$ (b) 85 (c) $(1010101)_2$ (d) Sum of place values = 85

10

A file has a size of 5 KB (where 1 KB = 1024 bytes and 1 byte = 8 bits). (a) Calculate the total number of bits in the file. (b) Express 1024 as a binary number. (c) If exactly one bit in the binary form of 1024 is flipped from 1 to 0, what are all the possible resulting decimal values?

Solution**Part (a) — Total Bits in 5 KB**

$$5 \text{ KB} = 5 \times 1024 \text{ bytes} = 5120 \text{ bytes}$$

$$5120 \text{ bytes} \times 8 \text{ bits/byte} = \mathbf{40,960 \text{ bits}}$$

Part (b) — Convert 1024 to Binary

$$1024 \div 2 = 512, \text{ remainder} = \mathbf{0}$$

$$512 \div 2 = 256, \text{ remainder} = \mathbf{0}$$

$$256 \div 2 = 128, \text{ remainder} = \mathbf{0}$$

$$128 \div 2 = 64, \text{ remainder} = \mathbf{0}$$

$$64 \div 2 = 32, \text{ remainder} = \mathbf{0}$$

$$32 \div 2 = 16, \text{ remainder} = \mathbf{0}$$

$$16 \div 2 = 8, \text{ remainder} = \mathbf{0}$$

$$8 \div 2 = 4, \text{ remainder} = \mathbf{0}$$

$$4 \div 2 = 2, \text{ remainder} = \mathbf{0}$$

$$2 \div 2 = 1, \text{ remainder} = \mathbf{0}$$

$$1 \div 2 = 0, \text{ remainder} = \mathbf{1}$$

$$\text{Binary} = \mathbf{1000000000}$$

Part (c) — Flipping Exactly One 1-bit to 0

$$1024 \text{ in binary} = 1000000000$$

There is only **one bit set to 1** — the leading bit at position 10.

$$\text{Flipping it to 0 gives: } 0000000000 = \mathbf{0}$$

All other bits are already 0, so flipping them to 0 produces no change.

Therefore, the only possible resulting decimal value is **0**.

Answer: (a) 40,960 bits (b) $(1000000000)_2$ (c) The only possible value is 0
